

Sheet Metal With 2 mm thickness

RAW MATERIAL PREPARATION

Material Procurement

- 2 mm thick metal sheet (pre-polished or raw)

Initial Sheet Cutting (Blank Size)

- Method: Water Jet Cutting
- Why: Cold cutting, no thermal distortion, high precision for sheet metal.
- Use: Cut the outer dimensions (base size of the sheet).

PROFILE & HOLE CUTTING

Outer Profile Contours

- Method: Water Jet Cutting or Wire EDM (for tighter tolerance)
- Use: Shapes, notches, or rounded corners.

Hole Cutting

- Method:
 - Wire EDM for precise holes
 - Laser Drilling or Plasma for rough holes (if allowed in "non-conventional")
- Use: Ø6 mm, Ø5.65 mm, or any slotted holes seen in drawing.

SURFACE MODIFICATION (OPTIONAL)

5. Edge Smoothing / Deburring

- Method: Abrasive Flow Machining or Vibratory Finishing
- Use: Smoothen edges post-cutting.

6. Surface Texturing or Polishing

- Method: Electrochemical Polishing
- Use: For stainless steel parts to improve corrosion resistance and appearance.

FINAL STEPS

Cleaning

- Method: Ultrasonic Cleaning
- Use: Remove micro-particles, water jet residue, or polishing fluid.

Inspection

- Tool: CMM or Laser Scanner
- Use: Dimensional checks for tolerances, hole position, and flatness.

SUMMARY TABLE

Step	Method	Equipment
Raw Sheet Cut	Water Jet	CNC Water Jet Cutter
Hole and Slot Machining	Wire EDM / Laser Drilling	Wire EDM / Laser Driller
Profile Cutting	Water Jet / EDM	CNC Water Jet or EDM
Edge Finishing	Abrasive Flow / Vibratory	Deburring Equipment
Surface Polishing	Electrochemical Polishing	Electro-polisher
Cleaning	Ultrasonic Cleaning	Ultrasonic Cleaner
Dimensional Inspection	CMM / Scanner	Coordinate Measuring Machine

Base

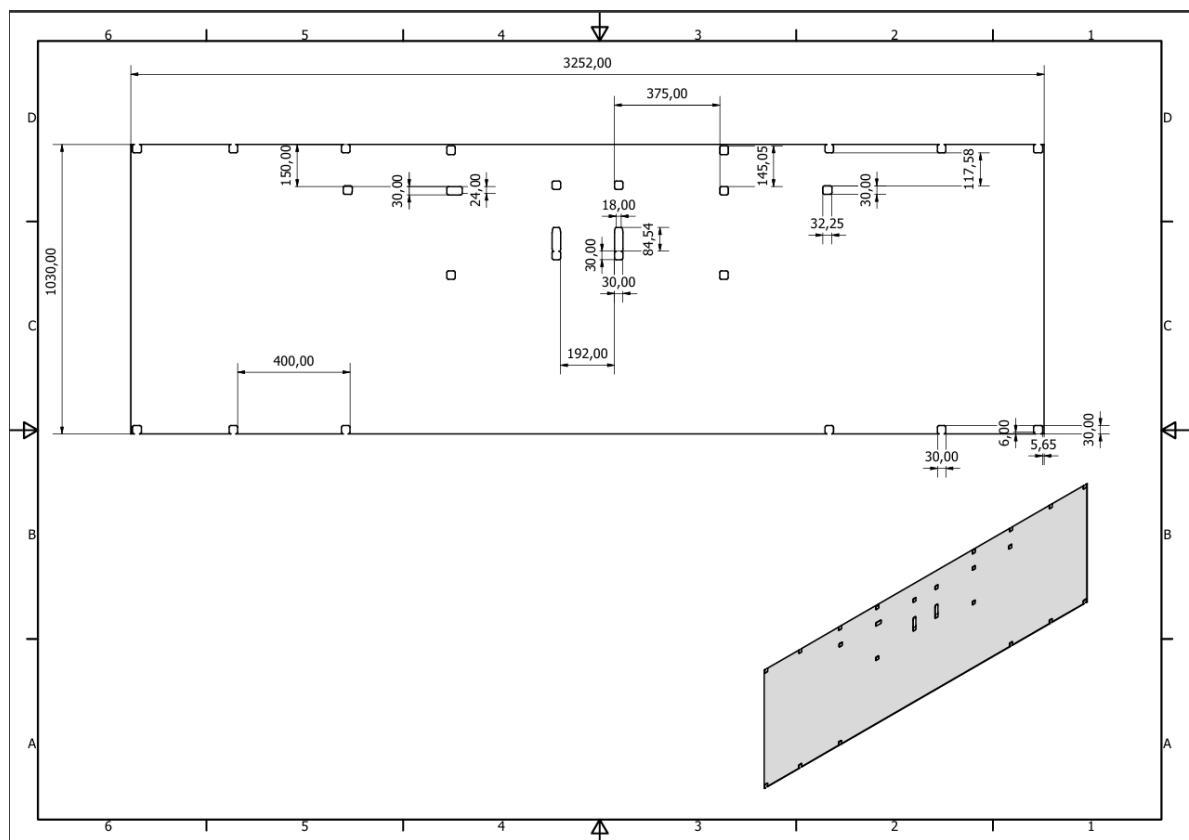


Figure 1 Base Sheet Working Drawing

Side

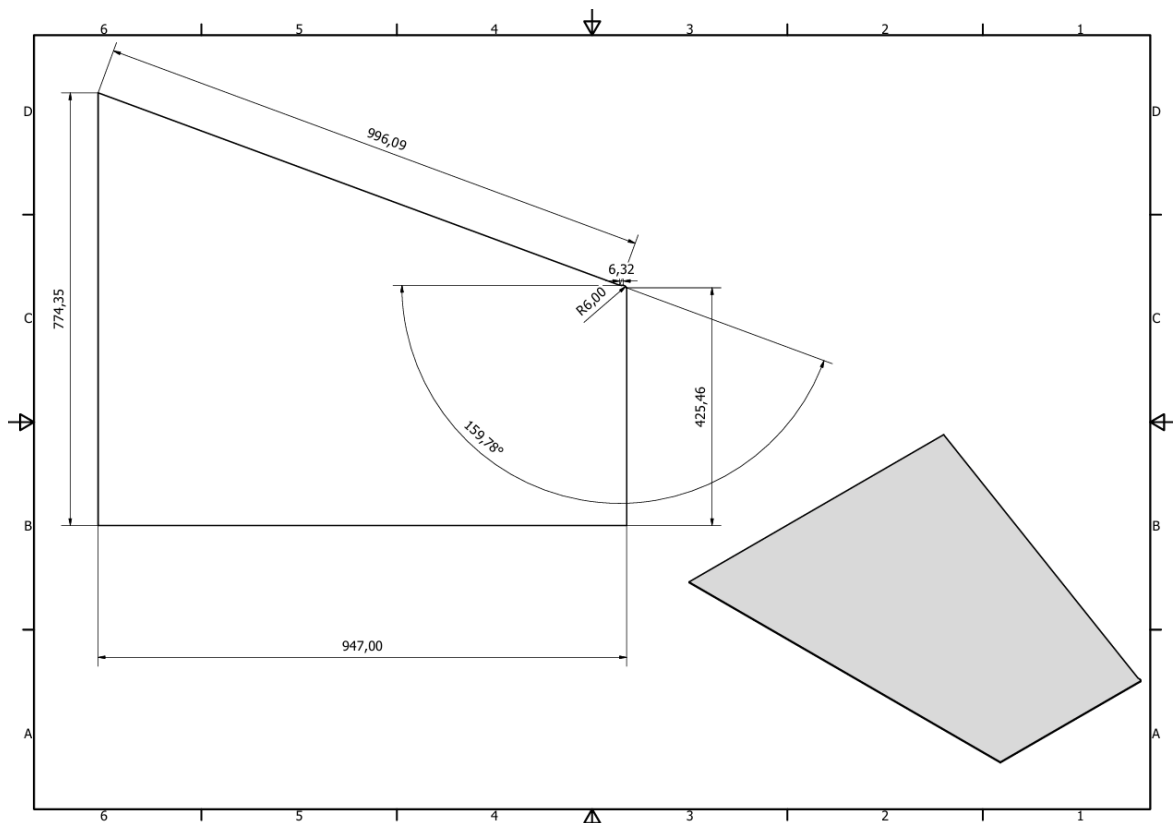


Figure 2 Side Sheet Working Drawing

Inclined 1

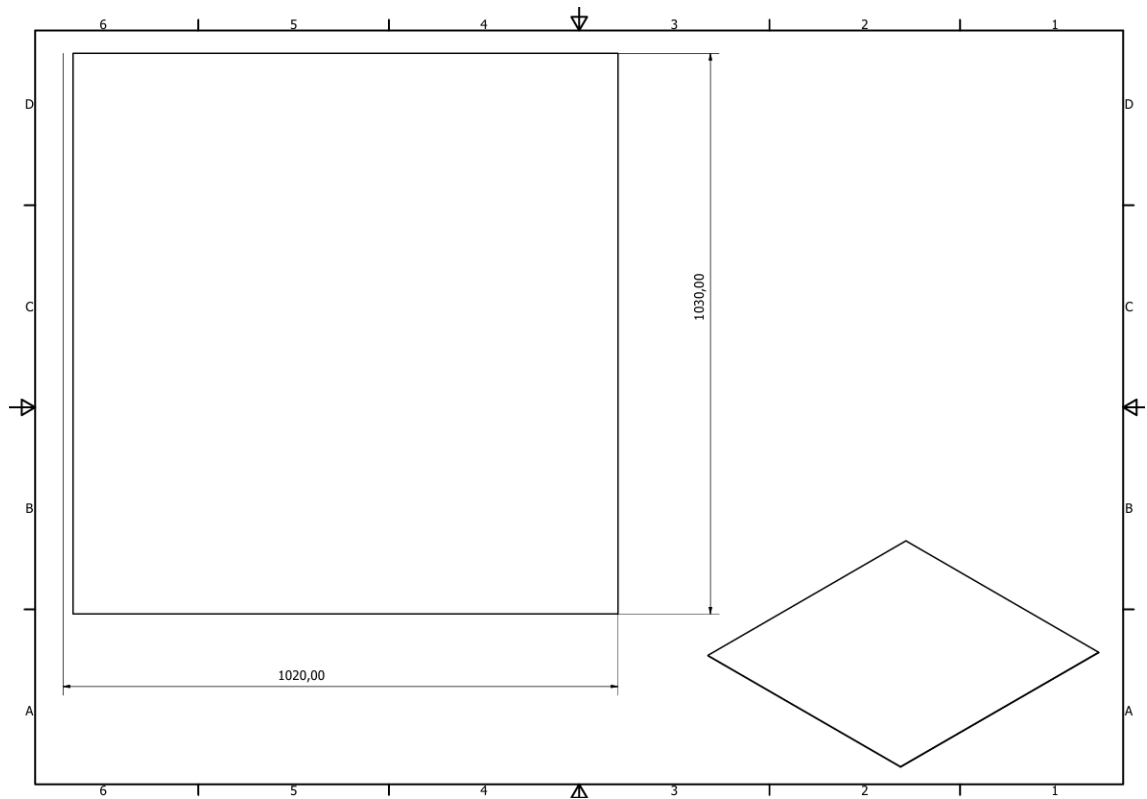


Figure 3 Inclined 1 Sheet Working Drawing

Mid Base 1

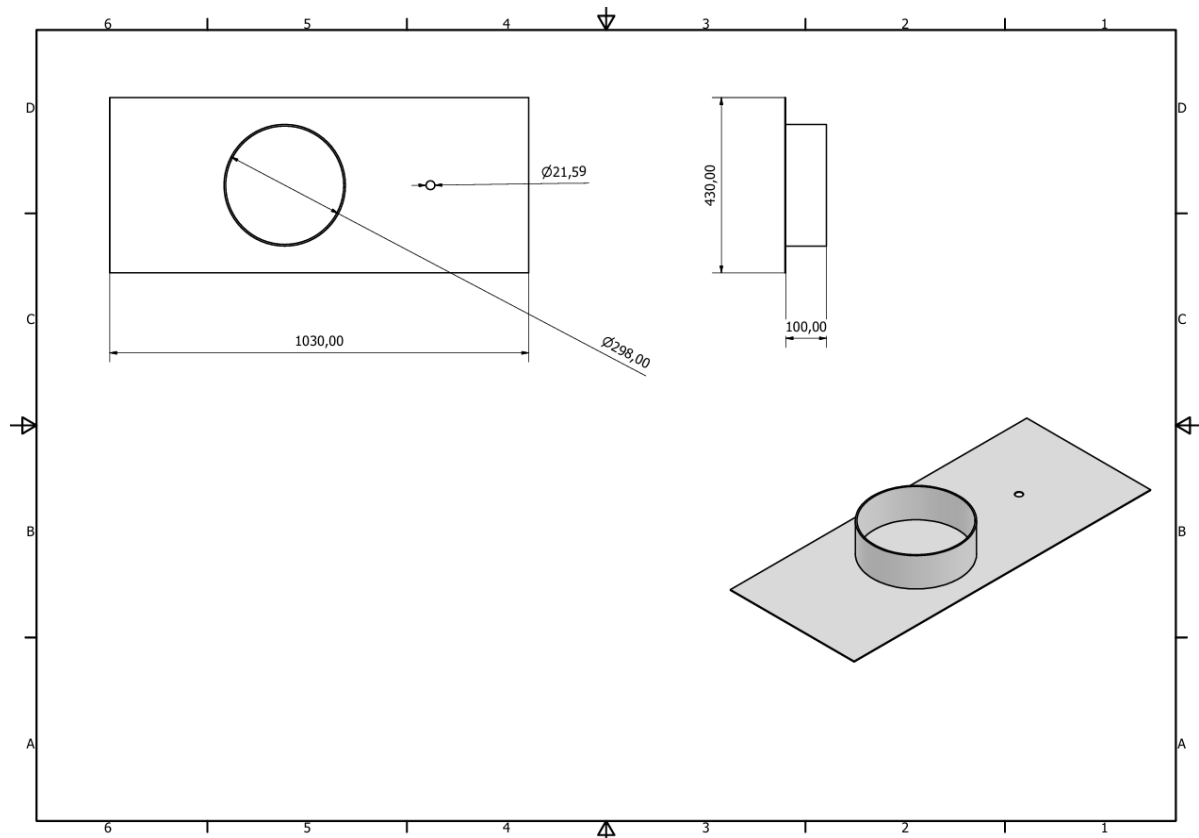


Figure 4 Mid Base 1 Sheet Working Drawing

Mid Base 2

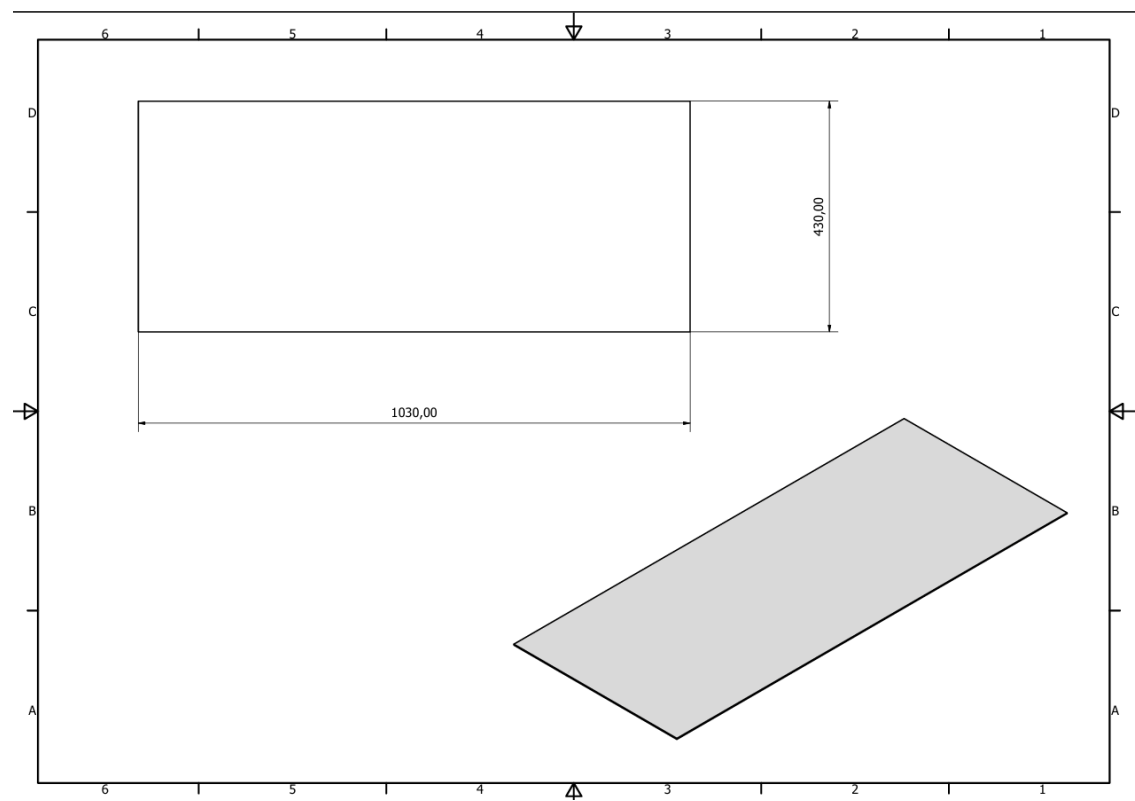


Figure 5 Mid Base 2 Sheet Working Drawing

Inclined 2

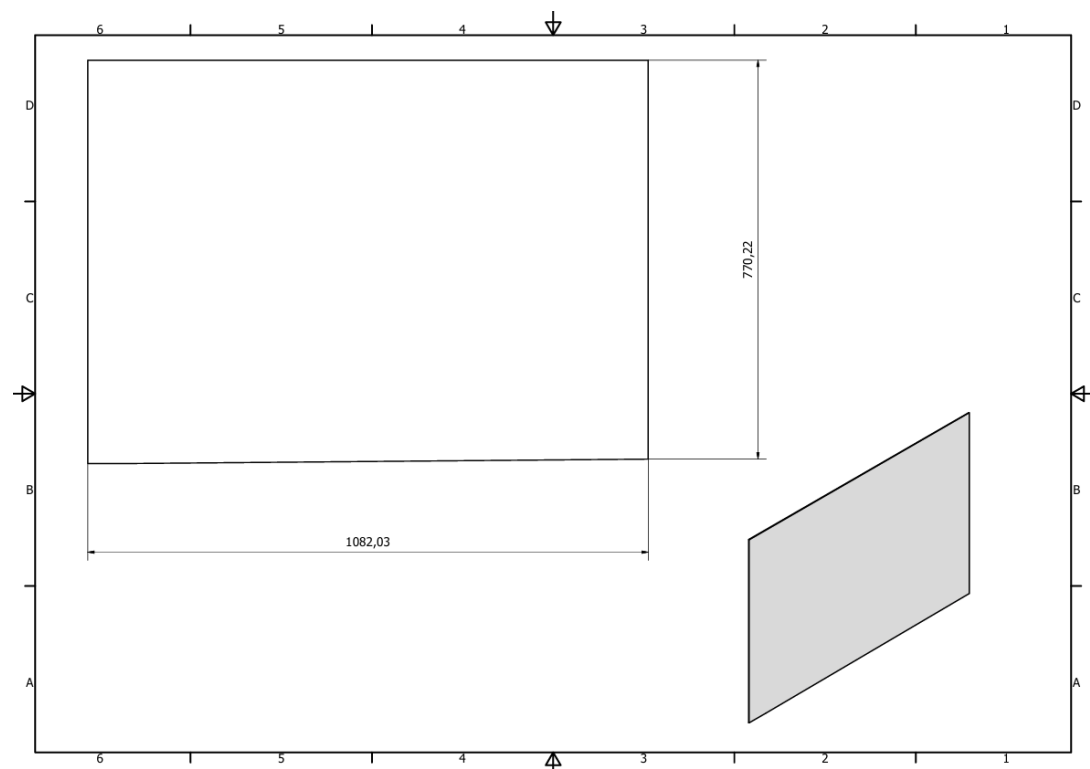


Figure 6 Inclined 2 Sheet Working Drawing

Top

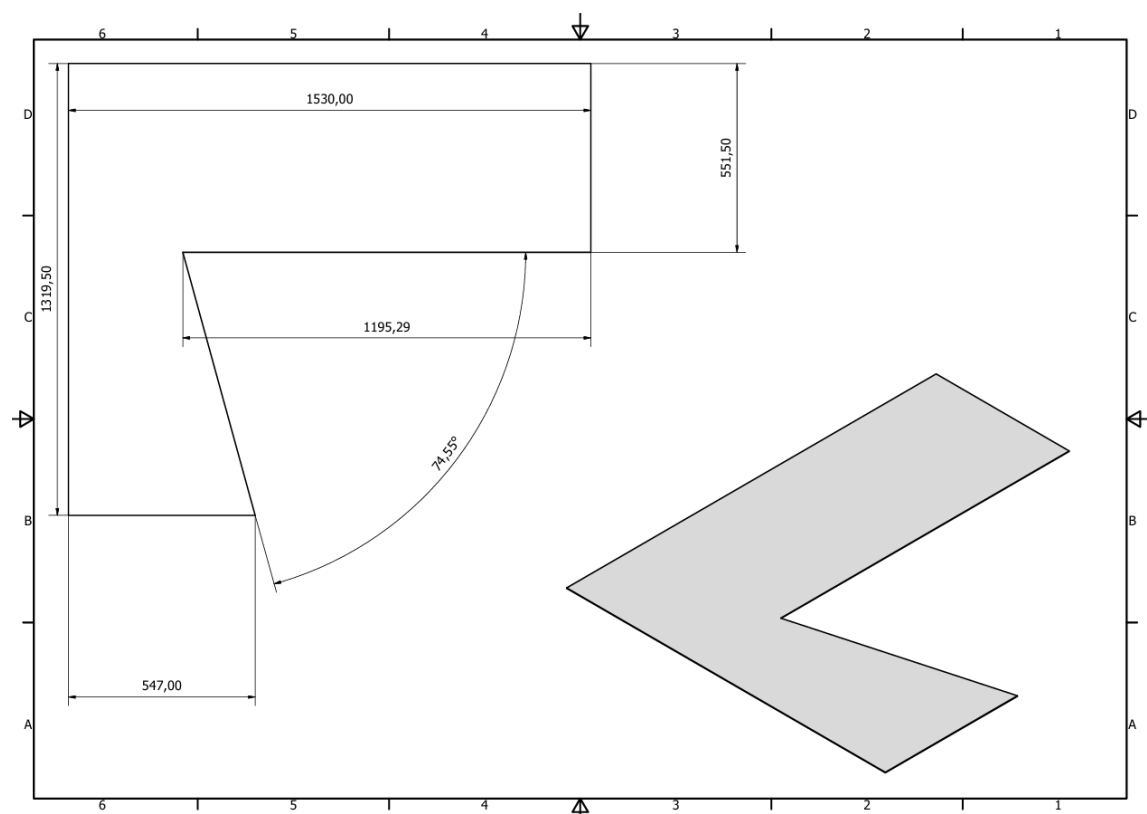


Figure 7 Top Sheet Working Drawing

Top Base

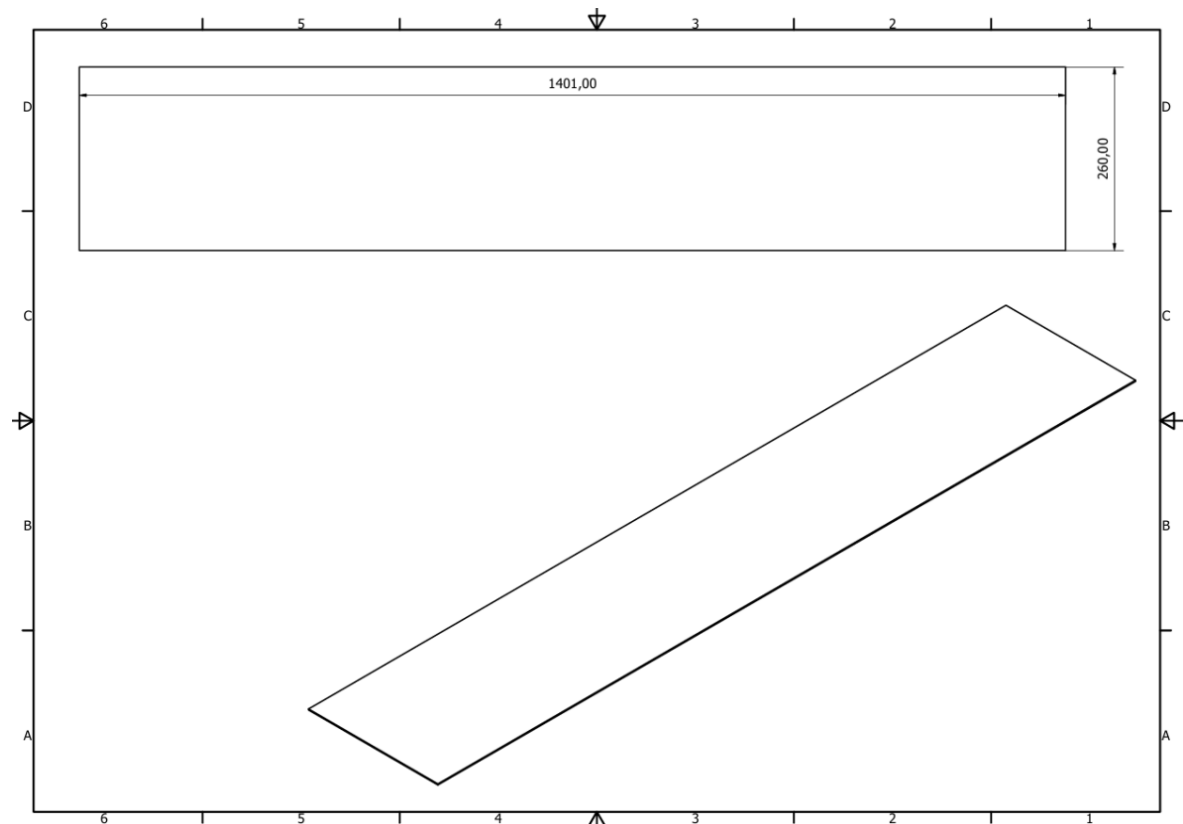


Figure 8 Top Base Sheet Working Drawing

Triangle

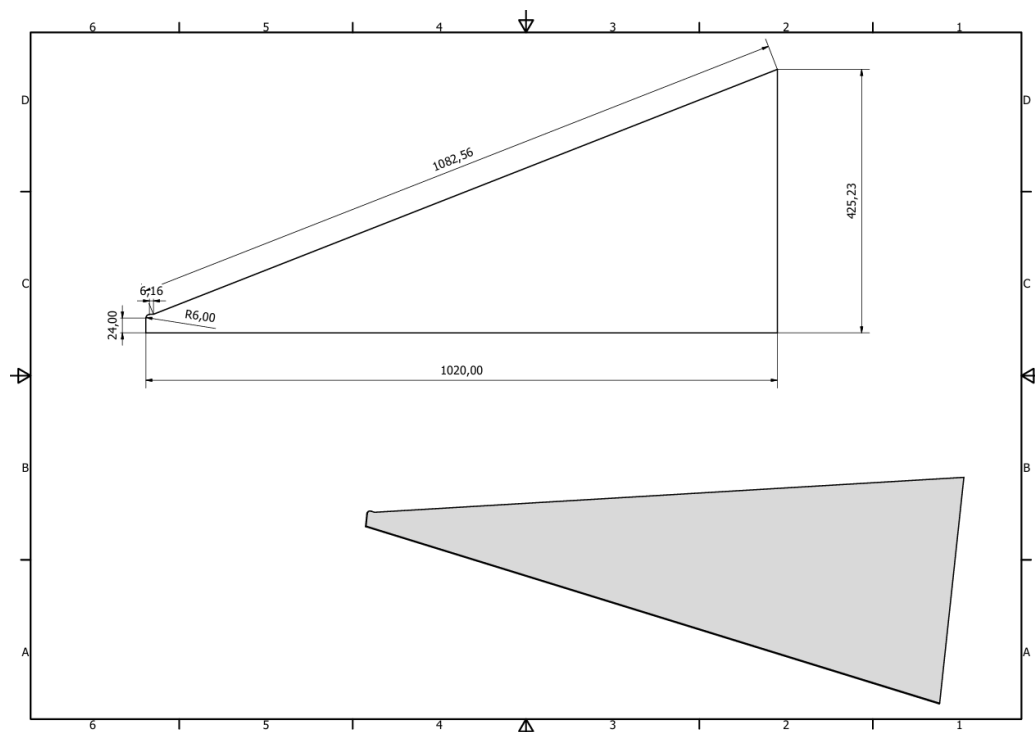


Figure 9 Triangle Sheet Working Drawing

Vertical

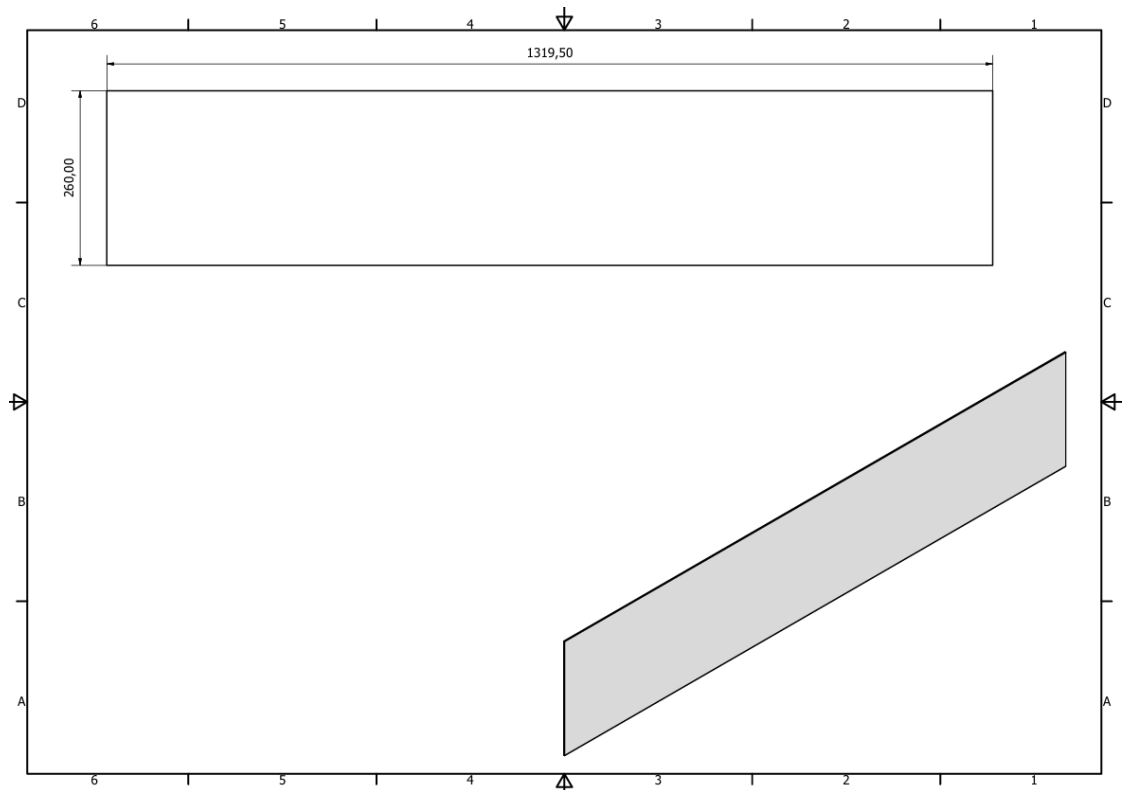


Figure 10 Vertical Sheet Working Drawing